

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 30)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0	1
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 A conservation team is planning to protect a population of an endangered antelope in a reserve.

0	1
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 .

1

 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size

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Distribution

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Survival rate

.....

.....

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Age structure

.....

.....

.....

0	1
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2

 Describe how the population size of the antelope could be estimated.

[2 marks]

.....

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.....

.....

10

0 2 . 1

Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

.....

Ozone depletion

.....

Tourism

.....

Future mineral exploitation

.....

0 2 . 2

Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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10

03 . 1

Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed

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Sea lice and disease

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.....

Escapes affecting wild gene pools

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Organic waste pollution

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03 . 2

Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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.....

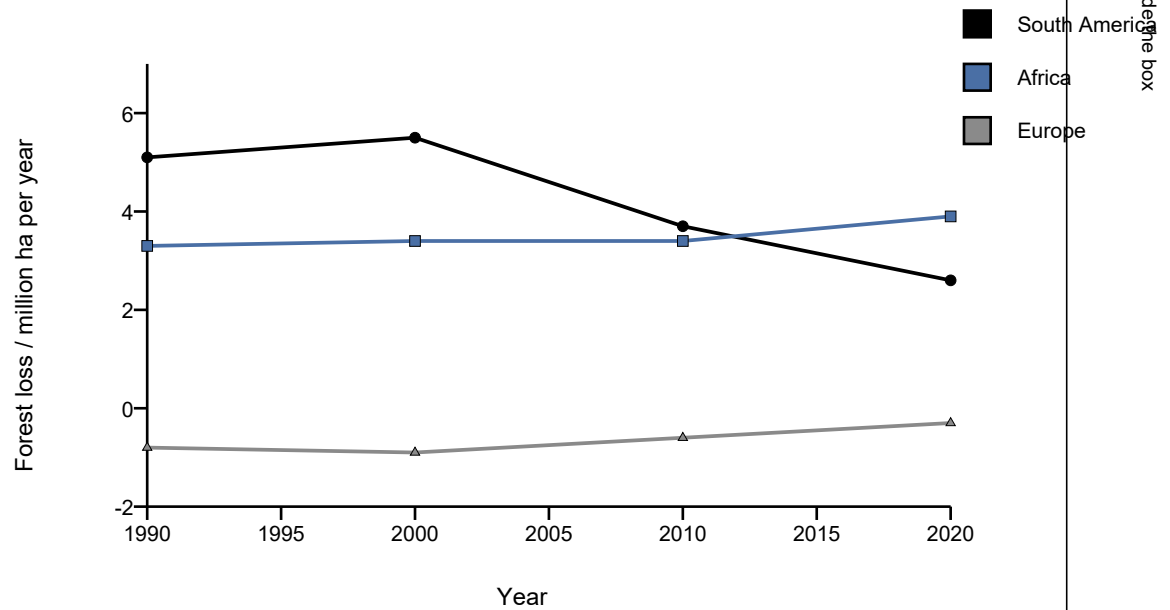
.....

10

0 4

Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 4

1 Describe the trends shown in Figure 1.

[3 marks]

0 4

2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

0 4 . 3

Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

0 4 . 4

Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

10

0 5 . 1

State what is meant by **hybrid vigour**.

[1 mark]

0 5 . 2

Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

0 5 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

2

5

0 6 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

.....

Soil maintenance

.....

Interspecies relationships

.....

0 6 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

.....

0 6 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

.....

10

07 . 1

Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique.

[6 marks]

Table 1

Organisms	Technique	One limitation	
Invertebrates active on the ground surface at night			
Invertebrates living in leaf litter			
Invertebrates on tree branches			

07 . 2

Explain why a light trap cannot be used to prove that a moth species is absent from a site.

[2 marks]

07 . 3

Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

[2 marks]

10

08 . 1

Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

08 . 2

Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

08 . 3

Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

08 . 4

Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

10

09 . 1

Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

09 . 2

Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

09 . 3

Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

09 . 4

Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

10

Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

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Extra space

10 . 2 State what is meant by **rewilding**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

Do not write outside the box

END OF QUESTIONS